

Supplementary Reference Paths and Evaluation Metadata

AD-STGN

These definitions accompany the frozen v3.6 evaluation. They are exported from `tep_meta_final.py`; the machine-readable CSV and JSON files are supplied alongside this document. The held-out target is XMEAS(35), represented at the unit level by PURGE.

S1. Ordered reference paths

The routes below define the reference unit sequences for ordered path recovery. They are task-specific evaluation annotations, not trajectories inferred by the model.

Source unit	Ordered reference route
FEED	FEED → REACTOR → SEPARATOR → PURGE
RCOOL	RCOOL → REACTOR → SEPARATOR → PURGE
SCOOL	SCOOL → SEPARATOR → PURGE
REACTOR	REACTOR → SEPARATOR → PURGE

S2. Unit-level process-influence reference graph

These 13 directed edges are used for distinct-edge evaluation. The category labels follow the frozen metadata. The graph includes indirect process influences and must not be interpreted as a list of immediate physical piping connections. A row denotes influence from the source unit to the target unit.

Source	Target	Category
FEED	REACTOR	direct material
REACTOR	SEPARATOR	direct material
SEPARATOR	COMPRESSOR	direct material
COMPRESSOR	REACTOR	direct material
SEPARATOR	PURGE	direct material
SEPARATOR	STRIPPER	direct material
SCOOL	SEPARATOR	energy coupling
RCOOL	REACTOR	energy coupling
SEPARATOR	REACTOR	recycle feedback
COMPRESSOR	PURGE	recycle feedback
REACTOR	PURGE	indirect propagation
FEED	COMPRESSOR	indirect propagation
RCOOL	STRIPPER	indirect propagation

The separately supplied `tep_search_unit_edges.csv` contains the ten allowed transitions retained in the search/support metadata. It is not interchangeable with the 13-edge evaluation graph.

S3. Fault scenarios and source-unit labels

Ten held-out realizations of each scenario form the 130-event evaluation partition. Source labels are defined at the physical-unit level.

Fault	Programmed disturbance	Source
IDV(1)	A/C feed ratio, B composition constant	FEED
IDV(2)	B composition, A/C ratio constant	FEED
IDV(3)	D feed temperature	FEED
IDV(4)	Reactor cooling water inlet temperature	RCOOL
IDV(5)	Condenser cooling water inlet temperature	SCOOL
IDV(6)	A feed loss	FEED
IDV(7)	C header pressure loss	FEED
IDV(8)	A/B/C feed composition	FEED
IDV(10)	C feed temperature	FEED
IDV(11)	Reactor cooling water inlet temperature (random)	RCOOL
IDV(13)	Reaction kinetics	REACTOR
IDV(14)	Reactor cooling water valve sticking	RCOOL
IDV(15)	Condenser cooling water valve sticking	SCOOL

S4. Provenance and conventions

Node indices in the sensor mapping are zero-based; XMEAS and XMV identifiers retain the original one-based numbering. XMEAS(35) is excluded from the 40 graph variables. Directed CSV edges use source-to-target ordering; adjacency matrices use row i , column j for the edge $j \rightarrow i$.

Process reference: J. J. Downs and E. F. Vogel. A plant-wide industrial process control problem. *Computers & Chemical Engineering*, 17(3), 245–255 (1993). [https://doi.org/10.1016/0098-1354\(93\)80018-I](https://doi.org/10.1016/0098-1354(93)80018-I).

The unit aggregation, ordered routes, and process-influence categories are the task-specific definitions supplied with this implementation. This export does not constitute additional independent physical validation.